

Dion Machado

437-263-9627 | macdion2007@hotmail.com | [LinkedIn](#) | [GitHub](#) | Mississauga, Ontario

PROJECTS

3D Space Themed Portfolio | React, TypeScript, HTML, Tailwind, Three.js, Framer-Motion

- Engineered a 3D interactive portfolio using Three.js, React, and WebGL, rendering procedurally-generated celestial bodies at a stable 60 FPS
- Built a custom camera and orbital motion system to simulate planetary movement and depth within the 3D scene
- Animated clickable project icons with Framer Motion, creating a smooth, game-like navigation experience across the portfolio

Arcane Spell Casting | React, Typescript, HTML, CSS, MediaPipe, TensorFlow.js, Pixi.js

- Developed a program that triggers custom VFX based on real-time hand positioning using the MediaPipe handtracker
- Trained a custom lightweight gesture-classification model in TensorFlow.js on 900+ real-time camera samples, achieving 80%+ validation accuracy
- Mapped recognized gestures based on vector calculations and a gesture classification confidence to a library of custom particle and lighting effects triggered dynamically in the browser

Keel - A CSV Budget Dashboard - In Progress | React, TypeScript, Node.js, Tailwind, Plotly.js, SQL, Postgres

- Designing a budget dashboard that parses downloaded CSV exports of income and expenses, and displays visual analytics with Plotly.js in order to get a better look at my finances
- Developing a web-scraping pipeline to cross-reference transaction vendors against scraped merchant data, automating category labeling and eliminating manual entry
- Architecting a PostgreSQL database schema to persist transaction and budget history, enabling scalable querying and long-term data storage

EXPERIENCE

Aquatics Lifeguard & Swim Instructor

June 2024 – Present

Malton Community Center

Mississauga, Ontario

- Instructed 30+ swim classes across multiple age groups, adapting teaching methods to individual skill levels and learning styles
- Designed structured lesson plans that improved skill retention for 100+ students, balancing individualized attention with group instruction
- Maintained safety and performance standards in a high-stakes environment, monitoring 20+ swimmers per session while executing lesson objectives

Data Fest Hackathon Participant

April 2026

University of Toronto

Toronto, Ontario

- Cleaned and analyzed a real-world hospital dataset of 1M+ records using Python and pandas, addressing missing values and inconsistent formatting
- Built a linear regression model to identify statistically significant predictors of breast cancer risk, surfacing key trends in the data
- Translated findings into a data-driven proposal to improve the safety of over 10,000+ patients

EDUCATION

University of Toronto - St. George Campus

Toronto, Ontario

Bachelor of Arts in Data Science, Minor in Computer Science

Sept. 2025 – Current

- **Relevant Coursework:** Calculus with Proofs, Data Structures, Statistics, Mathematical Expression and Reasoning, Linear Algebra

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, HTML5, CSS

Libraries, Frameworks and Runtimes: React, Tailwind, TensorFlow.js, Three.js, NumPy, Node.js

Tools: Git, GitHub, Vercel, Vitest, React Router

Databases: PostgreSQL